# Compound Interest and Effective Rate

*interest on interest!*

Quantitative Aptitude: Business Mathematics

**In simple words** *Each period, interest is added to the principal, so the next interest is earned on a bigger amount.*

## 1 Compound Interest (CI)

Interest calculated on principal plus interest already added in earlier periods. CI = Amount − Principal.

*Not the same as Amount!*

## 2 Effective Rate

The equivalent annual rate, compounded once a year, that gives the same final amount as the nominal rate compounded m times a year.

1. For the same rate and time, CI is more than simple interest when time is above one period.
2. Nominal rate is the stated yearly rate; effective rate shows the true yearly growth.
3. More compounding periods in a year means a higher effective rate.
4. If compounding is yearly, effective rate equals nominal rate.



*Swipe for the formula →*

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# THE CI FORMULAS

Compound Interest and Effective Rate

rate ÷ m, time × m

## Amount with compounding

$$A = P\,(1 + r/(100m))^{(mn)}$$

A = amount, P = principal, r = nominal annual rate in %, n = years, m = compounding periods per year

## Compound interest

$$CI = A - P$$

CI = compound interest, A = final amount, P = principal

## Effective annual rate

$$E = (1 + i/m)^{m} - 1$$

E = effective rate (decimal), i = nominal annual rate (decimal, 8% = 0.08), m = periods per year

**SAME RATE, DIFFERENT GROWTH!** Compounding more often within a year raises the effective rate above the nominal rate.

Swipe for a memory trick ⟶

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# STUDY TRICK

Compound Interest and Effective Rate

*divide the rate, multiply the time*

## RATE DOWN, TIME UP

Compounding m times a year: divide the annual rate by m and multiply years by m. Half-yearly: half the rate, double the periods.

## HOW TO ATTEMPT IT

1 Note compounding frequency m → 2 Convert rate to r/m per period →

3 Convert time to m × n periods → 4 Find A = P(1 + rate)^periods →

5 Subtract P if CI is asked

### EXAM TIP

Read the words "per annum, compounded half-yearly" first; options often include the yearly-compounding answer as a trap.

### DOUBT BUSTER

*Is the effective rate always higher than nominal?* Only when compounding is more than once a year. With yearly compounding, both are equal.

*Swipe for common mistakes* →

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# COMMON MISTAKES

Compound Interest and Effective Rate

*avoid these slips*

✗ Reporting A as the compound interest

✓ CI = A – P. Subtract the principal at the end.

✗ Using full annual rate for half-yearly compounding

✓ Use r/2 per period and 2n periods.

✗ Putting r = 8 inside the effective rate formula

✓ Use i = 0.08 in decimal, then E = (1 + i/m)^m – 1.

✗ Using time in years with a per-period rate

✓ Rate and time must be in the same period unit.

**TOMORROW'S CA FOUNDATION MCQ TESTS THIS!**

Revise it tonight, then try the MCQ.

*Save this for revision!*

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